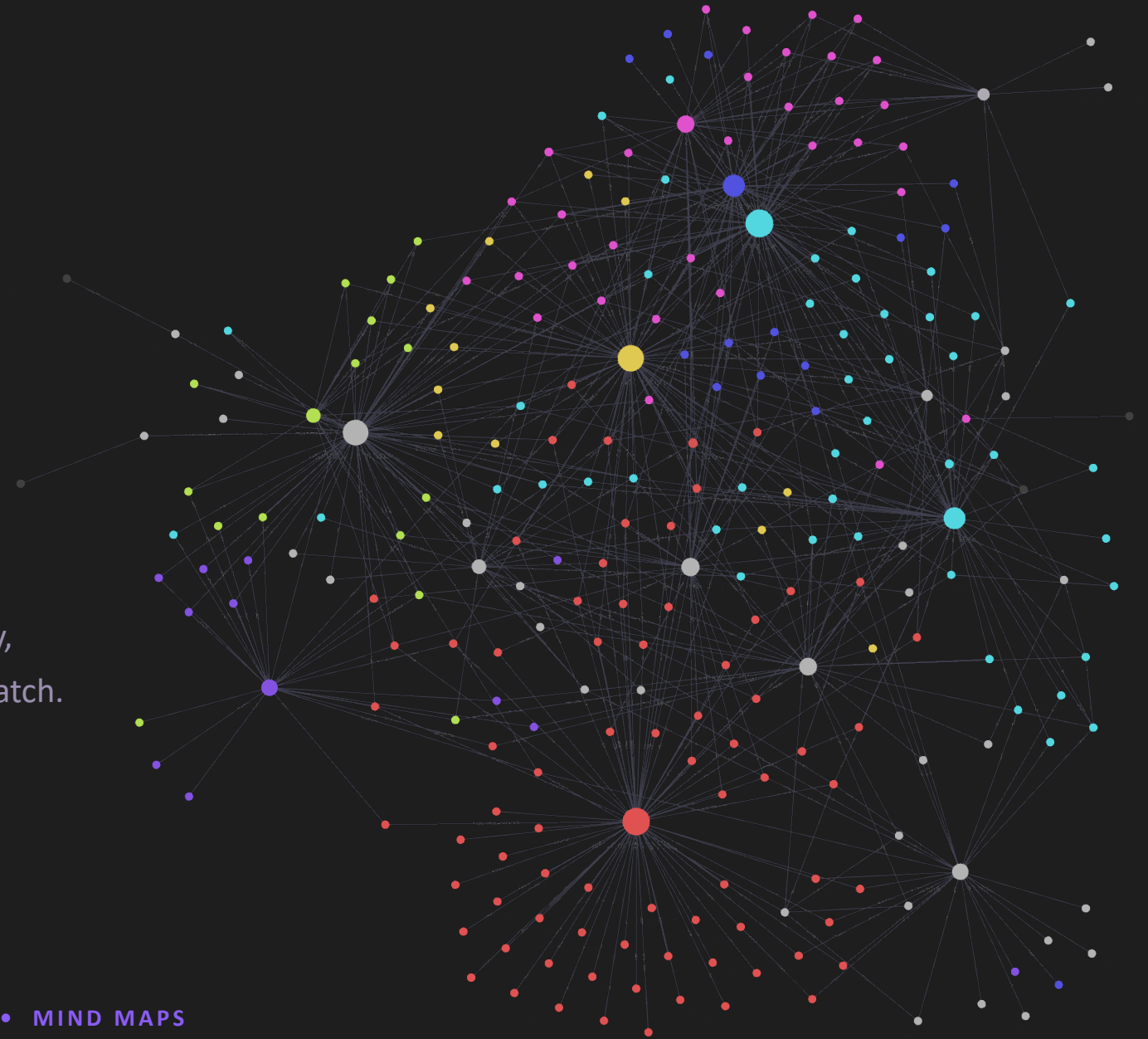


A BEGINNER'S GUIDE

Obsidian

A note-taking tool for research, science, bibliography,
and project & task management, explained from scratch.



PLAIN-TEXT NOTES • LOCAL FILES • LINKED THINKING • MIND MAPS

What is Obsidian ?

Obsidian is a free app for taking notes as plain-text Markdown files, stored locally on your computer. Instead of folders alone, you connect notes to each other with links - turning your notes into a personal, explorable web of ideas.



Markdown files

Notes are plain .md text files you fully own



Your computer

Stored locally in a folder called a "vault"



Free to use

Core app is free; optional paid sync/publish



1000+ plugins

Extend it for almost any workflow

THE BUILDING BLOCKS

RUN - mdp

```
integrator = md
dt = 0.02
nsteps = 300000000 ;#6 micro
nstxout = 0
nstvout = 0
nstfout = 0
nstlog = 50000
nstxout-compressed = 50000
nstenergy = 50000
compressed-x-precision = 10000
```

```
ntcomm = 100
comm-grps = system
comm-mode = linear
```

```
cutoff-scheme = Verlet
nlist = 20
ns_type = grid
rlist = 1.35
pbc = xyz
verlet-buffer-tolerance = -1
```

```
coulombtype = reaction-field
rcoulomb = 1.1
epsilon_r = 15 ; 2.5 (with polarizable water)
epsilon_rf = 0
vdw_type = cutoff
vdw-modifier = Potential-shift-verlet
rvdw = 1.1
```

```
tcoupl = v-rescale
tau_t = 0.1
```

equilibration - mdp
define = deflexible / dpoures

-DFLEXIBLE will use flexible water instead of rigid water into your topology, this can be useful for normal mode analysis.
-DPRES will trigger the inclusion of `psre.itp` into your topology, used for implementing position restraints.

How to use the Verlet scheme

The Verlet scheme is enabled by default with option `cutoff-scheme`. The value of `[.mdp] option verlet-buffer-tolerance` will add a pair-list buffer whose size is tuned for the given energy drift (in kJ/mol/ns per particle). The effective drift is usually much lower, as grompp assumes constant particle velocities. (Note that in single precision for normal atomistic simulations constraints cause a drift somewhere around 0.0001 kJ/mol/ns per particle, so it doesn't make sense to go much lower.) Details on how the buffer size is chosen can be found in the reference below and in the Reference Manual.

For constant-energy (NVE) simulations, the buffer size will be inferred from the temperature that corresponds to the velocities (either those generated, if applicable, or those found in the input configuration). Alternatively, `verlet-buffer-tolerance` can be set to -1 and a buffer set manually by specifying `rlist` greater than the larger of `rcoulomb` and `rvdw`. The simplest way to get a reasonable buffer size is to use an NVT mdp file with the target temperature set to what you expect in your NVE simulation, and transfer the buffer size printed by grompp to your NVE `[.mdp]` file.

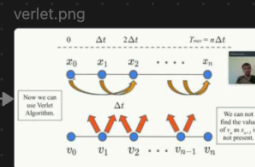


Figure 1A : pipeline

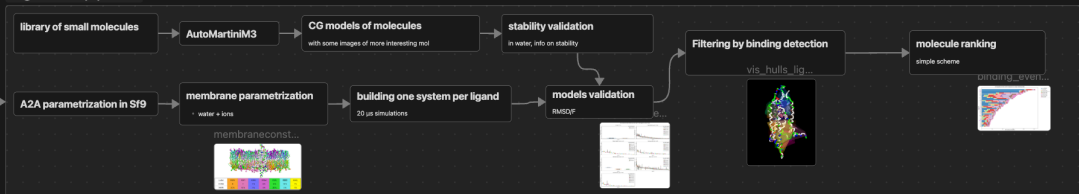


Figure 1

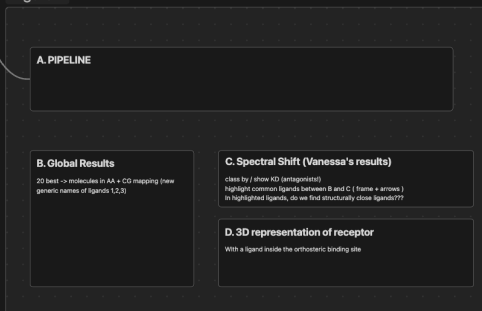


Figure 2

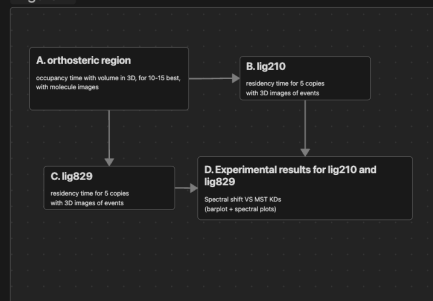
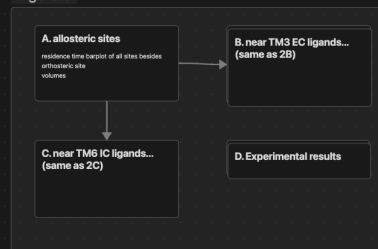


Figure 3



Vault

A folder on your computer holding all your notes. You can have several vaults for different areas of life.



Notes

Each note is a simple Markdown text file : readable in any text editor, forever, even without Obsidian.



Graph view

A visual map of every note and link - patterns and connections you didn't know were there become visible.



Links [[]]

Type `[[note name]]` to connect two notes. Links can be created before the target note even exists.

Bibliography with Obsidian



Connect ideas across papers



See backlinks automatically



Zoom out with the graph



Full-text search, instantly

The screenshot displays the Obsidian application interface. On the left is a sidebar with a file explorer showing a 'Bibliography' folder containing various categories like 'A2A', 'All-Atom MD', 'Biological Membranes', etc. The main area shows a note titled 'Gutierrez-2019'. The note content includes a title, year (2019), link, journal, authors (M. Gertrude Gutierrez, Jacob Deyell, Kate L. White, Lucia C. Dalle Ore, Vadim Cherezov, Raymond C. Stevens and Noah Malmstadt), categories (GPCRs, Protein-Lipid Interaction, Drug Research), and an abstract. The abstract describes the formation of giant unilamellar protein vesicles (GUPs) with the adenosine A_{2A} receptor (A_{2A}R) incorporated in the lipid bilayer. The bottom of the image shows a preview of the article 'The lipid phase preference of the adenosine A_{2A} receptor depends on its ligand binding state' from ChemComm, published on 30 April 2019.

Gutierrez-2019

Properties

- title: The lipid phase preference of the adenosine A_{2A} receptor depends on its ligand binding state
- year: 2019
- link: <https://doi-org.insb.bib.cnrs.fr/10.1039/c8cc10130b>
- journal: The Royal Society of Chemistry Communication

+ Add property

Authors:
M. Gertrude Gutierrez, Jacob Deyell, Kate L. White, Lucia C. Dalle Ore, Vadim Cherezov, Raymond C. Stevens and Noah Malmstadt

Categories:
GPCRs Protein-Lipid Interaction [[Drug Research]]

Abstract
Giant unilamellar protein vesicles (GUPs) were formed with the adenosine A_{2A} receptor (A_{2A}R) incorporated in the lipid bilayer and protein partitioning into the liquid ordered and liquid disordered phases was observed. When no ligand is bound, A_{2A}R partitions preferentially into the liquid disordered phase of GUPs, while ligand-bound A_{2A}R partitions into glycerol-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC) and cholesterol (chol) using a hydrogel-assisted method"

Commentary

PDF

ChemComm
COMMUNICATION
Published on 30 April 2019
Licence and permissions

The lipid phase preference of the adenosine A_{2A} receptor depends on its ligand binding state†

M. Gertrude Gutierrez,^a Jacob Deyell,^a Kate L. White,^{bcd} Lucia C. Dalle Ore,^b Vadim Cherezov,^{bcd} Raymond C. Stevens^{bcd} and Noah Malmstadt^{*,ade}

Cite this: Chem. Commun., 2019, 55, 5724
Received 21st December 2018.
Accepted 17th April 2019
DOI: 10.1039/c8cc10130b
rsc.li/chemcomm

Giant unilamellar protein vesicles (GUPs) were formed with the adenosine A_{2A} receptor (A_{2A}R) incorporated in the lipid bilayer and protein partitioning into the liquid ordered and liquid disordered phases was observed. When no ligand is bound, A_{2A}R partitions preferentially into the liquid disordered phase of GUPs, while ligand-bound A_{2A}R partitions into glycerol-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC) and cholesterol (chol) using a hydrogel-assisted method"

GPCRs are the largest family of proteins in the human genome.¹² They participate in intracellular signaling cascades and are implicated in behavioral and psychiatric disorders. In particular, A_{2A}R is implicated in addiction, in respiratory diseases such as chronic obstructive pulmonary disease (COPD) and is a

Bibliography with Obsidian



Connect ideas across papers



See backlinks automatically



Zoom out with the graph



Full-text search, instantly

The screenshot displays the Obsidian application interface. The main note, titled "GPCRs", contains a list of references including "positive allosteric modulators (PAMs)", "negative allosteric modulators (NAMs)", "neutral allosteric ligand (NAL)", "biased allosteric modulators (BAMs)", and "orthosteric". The left sidebar shows a file explorer with a "Bibliography" folder. The right sidebar shows a "Links" panel with a list of backlinks. A purple arrow points from the "Links" panel to the "Bibliography" folder.

Bibliography with Obsidian



Connect ideas across papers



See backlink



Zoom out with a wide view



Full-text search

Bibliography with Obsidian



Connect ideas across papers



See backlinks automatically



Zoom out with the graph



Full-text search, instantly

Time management

Daily notes

One note per day, created automatically from a template : journal, log, or planner.



Calendar view

See a month at a glance and jump straight into any day's note.



Templates

Pre-fill each day with goals, a task list, planned experiments and meetings



← →

_Templates / Daily

🔍 📅 ⋮

Daily

Links

- Meetings:
- Projects:

Todosy

Side notes

Results & figures

Sep 2026 < TODAY >

MON	TUE	WED	THU	FRI	SAT	SUN
31	1 ••	2 ••	3 •••	4 •	5	6
7 ••	8 •	9 ••	10 ••	11 •	12	13
14 ••	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4
5	6	7	8	9	10	11

Time management

Daily notes

One note per day, created automatically from a template : journal, log, or planner.



Calendar view

See a month at a glance and jump straight into any day's note.



Templates

Pre-fill each day with goals, a task list, planned experiments and meetings



09-09-2026

Links

- Meetings: [09-09-2026 WedCal](#)
- Projects: [Calixar Ligand Screen](#)

Todosy [12/15]

Simulations [4/4]

Rerun:

- ☒ ~~rerun replicata of APO 40µs system~~ ☒ 2026-09-09
- ☒ ~~duplicata HOLO Go Martini 20µs~~ ☒ 2026-09-09

Check:

- ☒ ~~check system HOLO with removed ligand in VMD~~ ☒ 2026-09-14
- ☒ ~~check APO 40µs system in VMD~~ ☒ 2026-09-14

Manuscript [4/5]

- ☒ ~~work on figure 3~~ ☒ 2026-09-09 [3/3]
- ☒ ~~3A : allosteric regions (occupancy time, 3D receptor + volume, mol img)~~ ☒ 2026-09-09
- ☒ ~~3B : five lig7 copies in allosteric regions (events snapshots above)~~ ☒ 2026-09-09
- ☒ ~~3C : five lig309 copies in allosteric regions (events snapshots above)~~ ☒ 2026-09-09

Sep
2026

< TODAY >

MON	TUE	WED	THU	FRI	SAT	SUN
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4
5	6	7	8	9	10	11

Tutorial on GitHub = your first vault



**CBI-PhDyingObsidianVault** Public

Unpin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

<> Code

**szczukam** update slides

362f082 · last week

19 Commits

obsidian	update slides	last week
Bibliography	update python script to remove tags and unpopular keywo...	2 months ago
Daily	first commit	2 months ago
Experiments	first commit	2 months ago
Meetings	first commit	2 months ago
Projects	first commit	2 months ago
Protocols	first commit	2 months ago
Results	first commit	2 months ago
_Attachments	add vis to readme	2 months ago
_Templates	first commit	2 months ago
.DS_Store	update python script to remove tags and unpopular keywo...	2 months ago
Obsidian_Guide.pdf	update slides	last week
README-short.md	update slides	last week
README.md	update slides	last week
zotero_to_obsidian.py	update python script to remove tags and unpopular keywo...	2 months ago

About

obsidian vault for a researcher in biology

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

**szczukam** Magdalena Szczuka

Languages

Python 100%

Suggested workflows

Tutorial on GitHub = your first vault

Table of contents

- 1. What is Obsidian?*
- 2. What is Markdown?*
- 3. Installing Obsidian and opening this vault*
- 4. The vault structure — what goes where*
- 5. Templates — what they are and how to use them*
- 6. How to create a note step by step*
- 7. How linking works — with a real example*
- 8. How to embed images and figures*
- 9. Recommended plugins and how to install them*
- 10. Daily workflow suggestion*
- 11. Bibliography management — importing from Zotero*



Tutorial on GitHub = your first vault

vault/	
└ Meetings/	one note per meeting
└ Daily/	one note per working day
└ Projects/	one note per research project
└ Protocols/	your experimental protocols
└ Experiments/	one note per experiment run
└ Results/	figures and graphical results
└ _Attachments/	images and files embedded in notes
└ _Templates/	templates (do not edit these)
└ README.md	quick reference guide

Meeting

Properties

📅 date dd/mm/yyyy
☰ people Empty
☰ project Empty
+ Add property

Meeting —

Date:
People:
Project: Projects/

Main topics

•

Notes

•

To do

☐

Ideas

-

Tutorial on GitHub = your first vault

vault/	
└ Meetings/	one note per meeting
└ Daily/	one note per working day
└ Projects/	one note per research project
└ Protocols/	your experimental protocols
└ Experiments/	one note per experiment run
└ Results/	figures and graphical results
└ _Attachments/	images and files embedded in notes
└ _Templates/	templates (do not edit these)
└ README.md	quick reference guide

Daily

Properties

📅 date

📅 dd/mm/yyyy

+ Add property

#

What I did

•

Experiments run

• [Experiments/](#)

Observations & issues

•

To do tomorrow

☐

Links

• Meetings:

• Projects:

Tutorial on GitHub = your first vault

vault/	
└ Meetings/	one note per meeting
└ Daily/	one note per working day
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└ Protocols/	your experimental protocols
└ Experiments/	one note per experiment run
└ Results/	figures and graphical results
└ _Attachments/	images and files embedded in notes
└ _Templates/	templates (do not edit these)
└ README.md	quick reference guide

Protocol

Properties

📅 date	📅 dd/mm/yyyy
☰ version	v1
☰ project	Empty
☰ validated	yes / no
+ Add property	

Protocol —

Project: [Projects/](#)

Version: v1

Validated:

Last updated:

Purpose

| What is this protocol for?

Materials & reagents

Item	Quantity	Supplier / Reference

Steps

- 1.
- 2.
- 3.

Critical steps & warnings

Getting started, step by step

1



Install & open

Download Obsidian (free) for your computer and open vault from folder (GitHub tutorial) or create new vault from scratch.



2



Write your first notes

Start plainly: one note per topic, paper, or day.



3



Start linking

As ideas repeat, connect them with `[[links]]` - your own tags, keywords, ideas. Structure emerges from your notes.



4



Use plugins

It's not necessary but very useful - start with Tasks, Calendar or Citations.

Getting started, step by step

PhDying Time of my Life
Workspace

+ New Task

Today

Inbox

Upcoming

Forecast

Projects

Working On

Tags

Flagged

Table

Review

Matrix

PROJECTS

- [[Calixar Ligand Screen]] 17
- [[Protein LLLs interactions]] 13
- Administration 5
- Bibliography 7
- Add Project

OTHER VIEWS

- Events
- Status
- Plan

407 Tasks

Matrix

Priority (High to Low)

Urgent & Important	Not Urgent & Important
20 tasks ☐ prepare and send to the supplier the ligands and their names + their structures. I will select the first 10% ligands with shortest lead times and those that are already bought. !!!! AllTasks L45	2 tasks ☐ check the efficiency of the lipid binding assay and if there is any correlation between the results. !!! AllTasks L19
☐ figure 2D : experimental results (MST) for lig210 and lig829 (KD) #project/ !!!! AllTasks L36	☒ rerun lipid binding analysis on smaller frame count cutoff -> 3 nanos #project/ AllTasks L2
Urgent & Not Important	Not Urgent & Not Important
0 tasks ☒ envoyer le compte rendu avec la nouvelle liste de ligands a tester + les diapos présentés 27-05-2026 WedCal L38	385 tasks ☐ https://www.biorxiv.org/content/10.64898/.743494v1.full AllTasks L6
☒ report on what I do / did before going on vacation -> before 01-07-2026 WedCal L26	☐ https://www.nature.com/articles/s42004-026-02059-7 AllTasks L7
☒ Rerun 20 micros A2A-Sf9 on Irene ...	☐ https://www.sciencedirect.com/science/article/pii/S0959440X26000308?via%3Dihub



Use plugins

It's not necessary but very useful - start with **Task Genius**, Calendar or Citations.

Pros



Free core app

Full note-taking features cost nothing; only sync/publish add-ons are paid.



You own your files

Plain Markdown files - no lock-in, no proprietary format, readable forever.



Works fully offline

No internet connection needed; everything lives on your device.



Deep customization

Community plugins and themes adapt it to almost any workflow.



Fast, local search

Instant full-text search across huge vaults, even with many PDFs attached.

Cons



Learning curve

Linking, plugins, and Markdown syntax take longer to learn than a simple notes app.



No built-in real-time collaboration

Not designed for co-editing a note live with others, unlike Notion or Google Docs.



Organization is on you

Few built-in rules - an unstructured vault can get messy without personal discipline.